

BLOCK Data Engineering - Unit on Feature Selection, Part 1

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FACULTY OF
COMPUTER SCIENCE



1 Feature Selection Overview

2 Correlations and Examples

3 More on 'goodness'

RECALL: on feature selection

- ▶ Feature selection workflows
- ▶ Goodness of features
- ▶ Correlations and
- ▶ 'what is a hypothesis'

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Tasks and instruments

- 1 Finding whether (two) features are *correlated*^a
- 2 Workflows for selecting *good* features
- 3 Quantifying goodness
- 4 and trusting the numbers

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Book chapters for this unit

Ch 7 'Feature Selection' & from Ch 3 'Data Preparation Basic Models'

- ▶ 3.2.1: Data Integration –'Finding Redundant Attributes'
- ▶ 3.5: Data Transformation

Interplay between set of features and amount of data

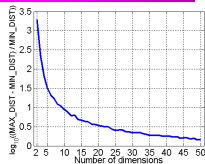
Two important aspects in the quest for features:

The more features we have, the more data we need.

Curse of Dimensionality

- When dimensionality increases, data becomes increasingly sparse in the space that it occupies

- Definitions of density and distance between points, which are critical for clustering and outlier detection, become less meaningful



• Randomly generate 500 points

• Compute difference between max and min distance between any pair of points

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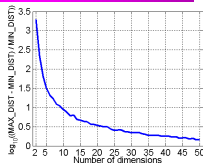
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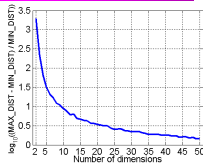
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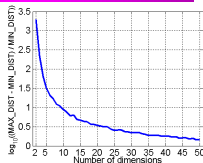
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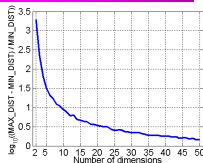
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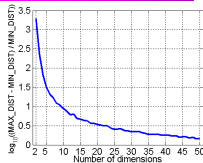
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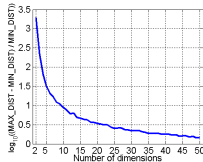
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All	479	4	N/A

"All" refers to patients that completed all the questionnaires

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Correlation between two *categorical* features

(χ^2 test)

Let A, B be two nominal (categorical) attributes with c , respectively r distinct values, in a dataset with m instances. **Informally**, we want to know whether the likelihood of observing the joint event $(A = a_i, B = b_j)$, i.e. $\frac{\text{count}(A=a_i \wedge B=b_j)}{m}$ is the same as $E[A = a_i] \times E[B = b_j] = \frac{\text{count}(A=a_i)}{m} \times \frac{\text{count}(B=b_j)}{m}$ – for all combinations of A and B values.

Formally:

χ^2 for two categorical attributes

(Section 3.2.1.1)

We compute

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{ij} is the *observed number of occurrences* of the joint event (A_i, B_j) , and

$e_{ij} = \frac{\text{count}(A=a_i) \times \text{count}(B=b_j)}{m}$ is the *expected number of occurrences*.

We test against the hypothesis that A, B are independent (H_0) with $(c - 1) \times (r - 1)$ degrees of freedom.^a

^aTo perform a statistical test like χ^2 we use a table or a software that can provide the values for the likelihood of the independence hypothesis.

Correlations by Example

START

The dataset 'Patients' and a question on two features

Id	I2	I15	I30	I22	I31	I9	I26	Response
#1	f	VH	yes	1	no	r	no	yes
#2	m	M	no	2	no	b	yes	no
#3	m	M	no	6	no	b	no	no
#4	f	VH	yes	7	no	b	no	yes
#5	m	L	no	3	no	l	no	no
#6	m	M	no	1	no	l	no	no
#7	f	VH	yes	1	yes	l	yes	yes
#8	f	H	no	2	yes	r	no	no
#9	f	H	yes	2	no	l	no	yes
#10	m	M	yes	6	no	b	no	no
#11	m	M	no	3	no	l	no	no
#12	f	H	no	4	no	r	no	no
#13	f	L	yes	4	no	l	yes	yes
#14	m	M	no	7	no	b	no	no
#15	m	L	no	5	no	l	yes	no

Are features I2 and I9 correlated?

STEP 1: For each value of I2 and each value of I9, count the joint occurrences

- ▶ I2=f and I9=r: 3
- ▶ I2=f and I9=b: 1
- ▶ I2=f and I9=l: 3
- ▶ I2=m and I9=r: 0
- ▶ I2=m and I9=b: 4
- ▶ I2=m and I9=l: 4

```
SELECT I2, I9, count(*)  
← FROM Patients  
GROUP BY I2, I9;
```


STEP 2: Bring all numbers in an overview table; build subtotals and crosscheck

		I9			Σ
		r	b	l	
I2	f	3	1	3	7
	m	0	4	4	8
Σ		3	5	7	15



STEP 3: Check whether we are permitted to perform the χ^2 test

IF Yes THEN goto STEP 4

ELSE ^a find another test

^ae.g. if the dataset is too small

We can now compute the likelihoods

Observed		Expected
I2=f and I9=r	3/15	$(7/15) * (3/15)$
I2=f and I9=b	1/15	$(7/15) * (5/15)$
I2=f and I9=l	3/15	$(7/15) * (7/15)$
I2=m and I9=r	0/15	$(8/15) * (3/15)$
I2=m and I9=b	4/15	$(8/15) * (5/15)$
I2=m and I9=l	4/15	$(8/15) * (7/15)$

! but how to compare ?

Assume that the Patients dataset is larger and the numbers look as follows:

STEP 1: count the joint occurrences

- ▶ I2=f and I9=r: 300 ▶ I2=m and I9=r: 0
- ▶ I2=f and I9=b: 100 ▶ I2=m and I9=b: 400
- ▶ I2=f and I9=l: 300 ▶ I2=m and I9=l: 400

STEP 2: overview table

		I9			
		r	b	l	Σ
I2	f	300	100	300	700
	m	0	400	400	800
Σ		300	500	700	1500

STEP 3: Check whether we are permitted to perform the χ^2 test

IF Yes THEN goto STEP 4

ELSE find another test

STEP 4: Compute **counts**

	Observed	Expected
I2=f and I9=r	300	$700 \cdot (300/1500)$
I2=f and I9=b	100	$700 \cdot (500/1500)$
I2=f and I9=l	300	$700 \cdot (700/1500)$
I2=m and I9=r	0	$800 \cdot (300/1500)$
I2=m and I9=b	400	$800 \cdot (500/1500)$
I2=m and I9=l	400	$800 \cdot (700/1500)$

... and the elements of the χ^2 formula

Nominator	Denominator
$(300 - 700 \cdot 3/15)^2$	$700 \cdot 3/15$
$(100 - 700 \cdot 5/15)^2$	$700 \cdot 5/15$
$(300 - 700 \cdot 7/15)^2$	$700 \cdot 7/15$
$(0 - 800 \cdot 3/15)^2$	$800 \cdot 3/15$
$(400 - 800 \cdot 5/15)^2$	$800 \cdot 5/15$
$(400 - 800 \cdot 7/15)^2$	$800 \cdot 7/15$

Semantics: for I2=f and I9=r (first row), we observed 300 occurrences and expected 140.

STEP 5: Compute the elements of the sum to the end

- ▶ $I2=f$ and $I9=r$: $\frac{(300-140)^2}{140} = 182.85$
- ▶ and alike $76.19 + 2.17 + 160 + 66.67 + 1.90$

suming up to approx 489.79.

STEP 6: Test against the H_0

- ▶ Compute the degrees of freedom: they are $(2 - 1)(3 - 1) = 2$
- ▶ Choose the significance level (aka α -level), eg set 0.01 (1%)
- ▶ Consult a table ^a to find the 'critical' χ^2 value for the α -level and the two degrees of freedom: it is 9.21

This means that 99 out of 100 times (significance level: 1%) the χ^2 for these data would have a value of 9.21 or less.

- ▶ Compare 9.21 to the computed value of χ^2 , i.e. 489.79:
- ▶ and decide whether the H_0 can be negated: this is the case here, 489.79 is much larger than 9.21.

^aThis is public domain. You search for it under names like 'chi square table', 'chi square critical value table' or 'chi square distribution table'.

Correlations by Example

END

When the two features are numerical

(Pearson ...)

Pearson product moment coeff for two numerical features (Section 3.2.1.2)

A, B with means $\bar{A}, \bar{B}$, standard deviations σ_A, σ_B :

$$r_{A,B} = \frac{\sum_{i=1}^m (a_i - \bar{A})(b_i - \bar{B})}{m\sigma_A\sigma_B} \in [-1, +1]$$

where (a_i, b_i) are the values of A, B for instance $i, i = 1 \dots m$.

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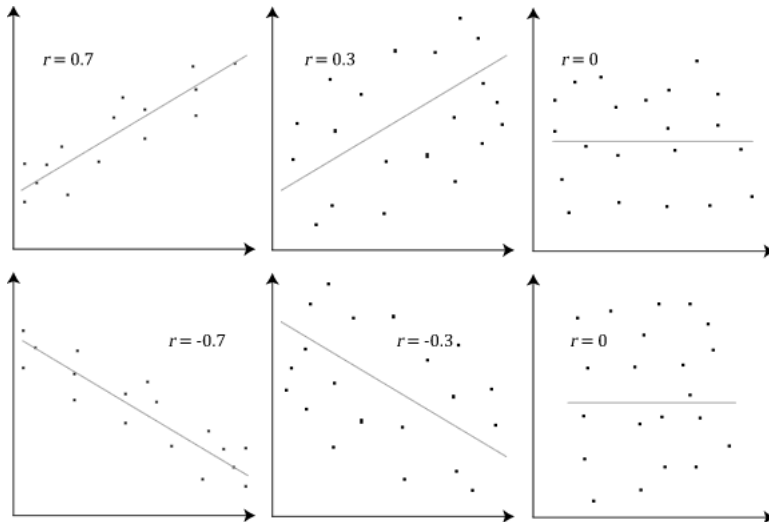
→

Pearson correlation coeff

(Section 7.2.2.3)

measures the degree of linear correlation between two variables X, Y with measurements $\{x_i\}$ and $\{y_i\}$ and means $\bar{x}, \bar{y}$:

$$\rho(X, Y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{(\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2)}}$$

Five values of $r_{A,B}$ EXAMPLE ¹¹Laerd Statistics, CC BY-SA 4.0<https://creativecommons.org/licenses/by-sa/4.0>, via Wikimedia Commons

When the features are numerical

Pearson & Covariance

Covariance between two numerical attributes (Section 3.2.1.2)

$$\text{Cov}(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{1}{m} \sum_{i=1}^m (a_i - \bar{A})(b_i - \bar{B})$$

- If A, B are independent, then $E(A \cdot B) = E(A) \cdot E(B)$, and thus $\text{Cov}(A, B) = 0$
- If A, B vary similarly, then whenever $A > \bar{A}$ we also expect that $B > \bar{B}$, and thus the covariance will be positive.
- If A, B vary in opposite directions, then whenever $A > \bar{A}$ we expect that $B < \bar{B}$, and thus the covariance will be negative.

²Covariance_trends.svg.png from Cmglee, CC BY-SA 4.0

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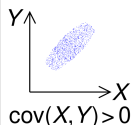
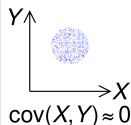
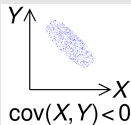
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When the features are ordinal (or ranks)

Spearman³

Spearman correlation coeff

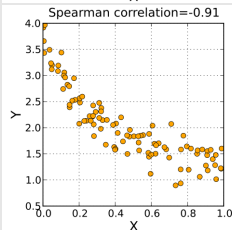
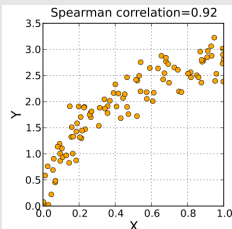
is the Pearson correlation coeff for the *ranks of the variables*, i.e.

$$r_s = \rho(R(X), R(Y)) = \frac{\text{Cov}(R(X), R(Y))}{\sigma_{R(X)} \sigma_{R(Y)}}$$

where X, Y are numerical variables and $R(X), R(Y)$ are their ranks.

Spearman correlation coeff can be used for ordinal variables, and, iff all ranks are distinct integers, then it holds that $r_s = 1 - \frac{6 \sum_{i=1}^m (x_i - y_i)^2}{m^3 - m}$ [Fieller et al., 1957]

Example



³Figures and quoted text accessed 2025-12-15 at

https://en.wikipedia.org/wiki/Spearman%27s_rank_correlation_coefficient

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⇒ one predicts the other

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! but this depends on how good the prediction is.

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- ⇒ one predicts the other
- ⇒ so it may suffice that only one is retained
 - ! but this depends on how good the prediction is.

...are correlated and one is *the target variable*:

- ⇒ the other one improves the prediction quality
- ⇒ so it should be retained
 - !! but this depends on how much the quality improvement is.

Goodness towards a target variable

Given a set of k classes/labels $\mathcal{C}$ with prior class probability $P(c_i)$ for each $i = 1, \dots, k$ and a feature space F :

- ▶ *Information measures*: What is the 'information gain' achieved when we consider a feature $A \in F$
 - (i) in comparison to $P(c_k)$ or
 - (ii) in comparison to considering another feature $B \in F$?(cf. Section 7.2.2.1)
- ▶ *Accuracy-like measures*: How does accuracy change when
 - (i) we consider $A \in F$ additionally to previous selected features,
 - (ii) we skip $A \in F$?(cf. Section 7.2.2.5)

Information gain

(Section 7.2.2.1)

The 'Information Gain' achieved by considering a feature A is the difference between the prior uncertainty $\sum_{i=1}^k U(P(c_i))$ and the expected posterior uncertainty using A :

$$IG(A) = \sum_{i=1}^k U(P(c_i)) - E \left(\sum_{i=1}^k U(P(c_i|A)) \right)$$

and as $U()$ we use Shannon's entropy (or a derived function), so that:

$$\sum_{i=1}^k U(P(c_i)) = - \sum_{i=1}^k P(c_i) \log_2 P(c_i)$$

and accordingly for the expected posterior uncertainty given A .

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What comes next: The feature selection process

Next time

Given a feature space F and a goodness measure: Build the subset of the good features S from scratch or rather by discarding features from F ?

Sequential forward feature set generation

Starting with an empty set, add one feature at a time, so that at each iteration the goodness improves.

Sequential backward feature set generation

Starting with the complete feature set, remove one feature at a time, the one that effects the least deterioration of goodness.

→ **The Quest for the Good Features** (Section 7.2.1.1, with modifications)

Thank you very much!

Questions?

Bibliography I



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